

1 BEFORE LEAVING THE OFFICE

SOP §5, §6

- ☐ Route, pass count and **direction** planned per pass
- ☐ **Satellite almanac checked** — gnssplanning.com/#/charts
- ☐ Two initialization locations identified — open sky, **both ends**
- ☐ GNSS-hostile stretches mapped; strategy for each
- ☐ Weather checked; stand-down criteria understood
- ☐ **Vehicle height clearances** checked on route
- ☐ Mechanical check — mounting, screws, torques
- ☐ System settings checked — lever arms, sensor settings
- ☐ **SSDs prepared** and field protocol printed
- ☐ Cleaning kit aboard · Wi-Fi password sticker accessible
- ☐ If air-freighted: **24 h acclimatisation complete**
- ☐ If idle > 2 months: run 30–60 min beforehand

2 SAFETY CHECK — BEFORE AND AFTER EVERY MISSION

SOP §6

Replace damaged parts immediately · tighten loose screws to correct torque · clean or dry anything wet

- ☐ Roof rack installed correctly; all screws tight
- ☐ Roof rack shows no cracks or deformation
- ☐ Sensor Unit damage-free — no scratches or deformation
- ☐ **All camera lenses clean and undamaged**
- ☐ Sensor in operation position and **properly locked**
- ☐ Control Unit undamaged, installed correctly, secured
- ☐ Power Unit undamaged, secured, **vents clear**
- ☐ All connectors plugged in and fixed
- ☐ Cables fixed to roof rack **before** entering the cabin
- ☐ No cable damaged or liable to be damaged
- ☐ Operator device working

DO NOT START a mission with an unresolved issue from a previous mission. Trimble: “Not doing this may damage the system permanently.”

3 POWER UP — ORDER MATTERS

SOP §7

- ① Engine **auto start/stop OFF**
- ② Both SSDs inserted and **locked**
- ③ **START THE VEHICLE**
- ④ Hold Control Unit power button **≥ 15 seconds**
- ⑤ Power Unit LED **constant green**
- ⑥ SU + CU LEDs blink ~10 s → **solid green**
- ⑦ Connect tablet → Chrome → <http://tmi.mx-scan.net>
- ⑧ Wi-Fi country set (first use / new country)

LED	Meaning
Blinking green	Starting / updating / shutting down
Solid green	Ready
Blinking red	Component failed — do not start a mission

WHY THE ORDER Startup draws **25 A**. Powering up on a non-running engine is how you meet the Battery Protect cutoff the hard way.

4 SYSTEM HEALTH BEFORE INITIALIZING

SOP §7

- ☐ Disk check complete — **no Warning or Error icon**
- ☐ Correct **Vehicle Preset** — install height matches this vehicle *today*
- ☐ **DMI / GAMS checkboxes ACTIVATED** if fitted
- ☐ Correct **Capture Preset** selected
- ☐ Camera and Laser buttons **GREEN**
- ☐ UTC time showing · vehicle parked in open sky

SILENT FAILURE An **unactivated DMI or GAMS logs nothing** — even though it is fitted, measured and connected. Nothing in the field will tell you.

5 INITIALIZATION — THE MOST IMPORTANT 20 MINUTES

SOP §8

- 1 Parked **open sky**, good PDOP, clear of buildings
- 2 Mission started — name, area, **presets confirmed**
- 3 **STILL for 2–3 minutes** — it feels long; it is not optional
- 4 Blue arrow visible; UTC time showing
- 5 **Drive straight ~20 m** → NAV **RED** → **ORANGE**
- 6 **Vary speed + 2–3 dynamic turns** → NAV **ORANGE** → **GREEN**
Example: 0 → 50 → 20 → 50 → 20 km/h
- 7 All parameters green in **Nav View**
- 8 **WAIT — up to 10 more minutes** before logging
- 9 Initialization time and location **recorded**

NAV	Meaning	Can you record?
RED	No valid solution / navigation failure	Blocked by the system
ORANGE	A solution exists	Permitted — but not for survey grade
GREEN	Meets your user accuracy figures	Yes — this is the target

FIELD TIP Watch *which* parameter holds you at orange. Heading → more dynamic manoeuvres. Position → move to a better location; more circles won't help.

6 NAVIGATION LOGGING VS. RECORDING — THREE RULES

SOP §8, §9

- Starts automatically** As soon as a valid position exists and the blue arrow appears — **even if NAV is still red**
- Unaffected by colour** Orange ↔ green transitions do **not** change navigation logging
- Stops on mission close** Closing the mission ends navigation logging **immediately**

The **Record** button starts and stops **laser and imagery only**. The trajectory records throughout the whole mission, including between runs.

7 COLLECTING

SOP §9

Record button

GREEN → press	Starts a run
RED while running	Logging is active
RED → press	Stops the run

Counter-intuitive: **red means recording**.

Driving

Recommended max speed	80 km/h (50 mph)
Absolute max speed	110 km/h (68 mph)
Minimum mission time	30 minutes
Style	Smooth — no harsh accel, braking or steering

DYNAMICS BEFORE, SMOOTHNESS DURING Initialization *wants* dynamics — they make errors observable. Collection wants the vehicle predictable. Every unmodelled jolt is a demand on the IMU and noise in the cloud.

8 MONITOR IN A LOOP WHILE DRIVING

SOP §9

Watch	Good	Act if
NAV button	GREEN	Orange — note the location · Red — recording is blocked
Camera / Laser	GREEN	Orange (not synced) or red (device error)
Record button	RED on a run	Green when you believe you are recording
Thick blue line	Over every collected stretch	Missing where you drove a run — the run is not recording
SSD fill status	Space remaining	Approaching full
UTC time	Displayed	Missing — GNSS initialization lost
Disk icons	None	Warning or Error icon appears
Audible alarm	Silent	Battery Protect — 78 s to restore charge before power cuts

BEST IN-VEHICLE QC CHECK TMI draws the trajectory as a **thin** blue line and recorded areas as a **thick** blue line. Every stretch you meant to collect should be thick. Finding a thin one here costs minutes; finding it in the office costs a mobilisation.

9 USEFUL DURING A MISSION

SOP §9

- | | |
|---|---|
| <ul style="list-style-type: none"> ☐ Comments — time-tags notes into the mission database. Use it. ☐ Waterfall view — confirms the laser is producing sensible data ☐ Camera exposure — adjust live when lighting changes materially | <ul style="list-style-type: none"> ☐ Record only project-specific areas ☐ Stop recording for turnarounds; restart on the new pass ☐ Need different capture settings? Use mission re-configuration — do NOT close the mission (no re-initialization needed) |
|---|---|

10 ENDING THE MISSION

SOP §10

- 1 Stop recording the final run
- 2 Drive to an initialization point, **dynamic manoeuvres on the way**
- 3 **FINALIZE SEQUENCE — mirror of the start:** dynamic steering → vary speed → drive straight → **STILL 2–3 min**
- 4 **Review the map** — thick blue line over every intended stretch
- 5 Close mission → **Complete Mission**
- 6 Shut down → **Shutdown**
- 7 Wait for power button light **OUT** — up to 90 seconds
- 8 Field protocol complete — runs, directions, date, mission, system S/N
- 9 Post-mission safety check (block 2)

NEVER PULL POWER Always use **Complete Mission** AND **Shutdown**, or risk corrupted data. **Do the map review BEFORE closing** — if a run is missing and you are still on site and initialized, you can re-drive it.

WHY THE MIRROR Office software solves the trajectory forwards from the start and backwards from the end, then blends them. A good initialization at *both* ends means the weakest part — the middle — is solved from two strong anchors instead of one. **It cannot be added later.**

11 DATA HANDLING

SOP §11

- ☐ System fully shut down before removing disks
- ☐ **Unlock and remove BOTH SSDs** — they are one dataset
- ☐ Offload via both Data Carrier Docks, USB 3
- ☐ **Verify:** navigation data · both lasers · panoramic · back-down camera · Mission Report · logs
- ☐ **Back up before any processing**
- ☐ **Only then** prepare SSDs for the next mission

TWO HARD RULES Never connect USB while a disk is in the Control Unit — remove it first. **Never delete raw mission data** because processing succeeded or the deliverable shipped.

12 STOP-WORK TRIGGERS

Blinking red LED	Do not start / stop
Disk Error icon	No production mission
Battery Protect alarm	78 s to restore charge
NAV red, material extent	Nothing is recording
Rain or mist	Trimble: avoid operating
Unresolved prior fault	Do not start

13 SUPPORT

SOP §14

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Provide:

- ☐ Problem description & how to reproduce
- ☐ Mission location and conditions
- ☐ **System log file** · serial number
- ☐ Total operating hours · photos / video

Log file is auto-saved to **removable Disk 1** during shutdown, or TMI.AI → Log Download.

Sources: Trimble MX60 User Guide Rev B (May 2025) · MX60 Quick Start Guide Rev B (March 2025) · TMI Software User Guide Rev L (April 2026) · Product Bulletin: Dust Filter in TMI for MX60 (January 2025). Page-level citations are in the numbered SOP sections.